



Continuous Assessment Test I – February 2024

Programme	: B.Tech (ECE/ECM)	Semester	: WS 2023-24
Course	: Electronic Materials and devices	Code	: BECE201L
		Class Nbr	: CH2023240501121 CH2023240501123
Faculty	: Dr. N.Chandrasekar Dr. Arivarasi A	Slot	: D2+TD2
Time	: 90 Minutes	Max. Marks	: 50

Answer ALL the questions

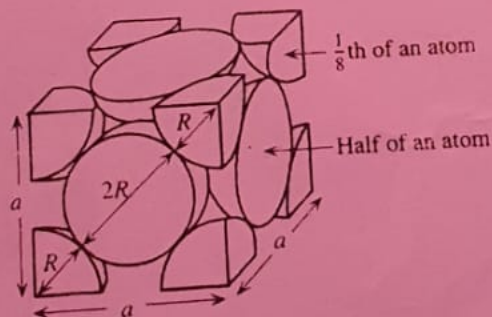
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Questions

Marks

- (i) Identify the crystal structure shown in Fig. 1. How many atoms are there per unit cell?
- (ii) Determine the relation between atomic radius and the lattice parameter.
- (iii) Calculate the Atomic Packing Factor (APF).
- (iv) Calculate the density of the crystal given that the atomic mass is 63.55 gm/cm^3 and the radius of the atom is 0.128 nm .

1.



[2+2+2+4]

Fig. 1

Electron drift mobility in Indium has been measured to be $6 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$. The room temperature (27°C) resistivity of In is $8.37 \times 10^{-8} \Omega \text{ m}$, and its atomic mass and density are 114.82 amu or g mol^{-1} and 7.31 g cm^{-3} , respectively.

- (i) Based on the resistivity value, determine how many free electrons are donated by each 'In' atom in the crystal.
- (ii) If the mean speed of conduction electrons in 'In' is $1.74 \times 10^8 \text{ cm s}^{-1}$, what is the mean free path?

[5+5]

3.

Germanium is doped with 5×10^{15} donor atoms per cm^3 at $T = 300 \text{ K}$. The dimensions of the Germanium sample used in Hall device are $d = 5 \times 10^{-3} \text{ cm}$, $W = 2 \times 10^{-2} \text{ cm}$,

[3+3+4]

and $L = 10^{-1}$ cm. The current is $I = 200 \mu A$, the applied voltage is $V = 100$ mV, and the magnetic flux density is $B = 5 \times 10^{-2}$ Tesla. Calculate:

- (i) the Hall voltage,
- (ii) the Hall field and
- (iii) the carrier mobility.

(i) Show that in Fermi-Dirac distribution, the probability that an electron occupies the Fermi level, E_F is always $1/2$ at any temperature.

4. (ii) Calculate the probability that a quantum state in the conduction band at $E = E_c + kT/2$ is occupied by an electron. The Fermi energy level is 0.25 eV below the conduction band. [3+7]

(i) Explain the concept of effective mass of an electron.

(ii) Explain how electron transitions occur in the indirect semiconductors.

(iii) The density of states near the band-edge in conduction band is given as [2+3+5]

5. $g_c(E) = \frac{4\pi(2m_e^*)^{3/2}}{h^3} \sqrt{(E - E_c)}$. Determine the number of states per unit volume in the range from E_c to $E_c + kT$ at $T = 300$ K. Given is $m_e^* = 1.08m_e$.

